Sagar Gupta

ProServe (Cloud Consultant) - DevOps/MLOps | Amazon Web Services (AWS) | Hyderabad, India

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Summary

Software Engineer with a strong algorithmic foundation – LeetCode Guardian (2166 peak, top 1.1%, 1300+ problems, 107 contests), Google Kick Start Round E '22 Rank 1289, 500+ GfG problems, HackerRank 6-Star Problem Solving. Two years at AWS Professional Services as ProServe (Cloud Consultant) - DevOps/MLOps, shipping DevOps platforms for enterprise customers and an MLOps pipeline AWS published as a sample. Deep across Python, TypeScript, React, FastAPI, AWS (SageMaker, ECS, Lambda), Terraform, and AI-lab-adjacent tooling (Claude Code plugins, MCP servers on Bedrock, Feast, Airflow). 11 upstream PRs merged (axios, Airflow, lucide, Prefect, Feast, awslabs/mcp, multer, and others). 5x AWS certified.

Experience

Amazon Web Services (AWS) Aug 2024 – Present
ProServe (Cloud Consultant) - DevOps/MLOps Hyderabad, India

- **Lead DevOps Consultant, Enterprise customer (Feb 2026 – Present):** Built a Terraform landing zone: Organizations OUs, Control Tower, SCPs, tag policies, account baselines and least-privilege deploy roles.
  - Automated network onboarding and tag-based firewall access with Transit Gateway.
  - Automated account vending with Terraform Cloud and OIDC federation.
- **DevOps Consultant, Global financial services customer (Oct 2024 – Aug 2025):** Wrote tested Terraform modules (terraform test, PyTest) for a multi-account landing zone. Worked on preventative security controls and EventBridge-based cross-account automation.
- **DevOps Consultant, Financial market infrastructure customer (Aug 2025 – Nov 2025):** Modernized legacy Terraform to cloud-native IaC patterns.
- **MLOps Pipeline (SME program, Jun – Aug 2025):** Built end-to-end SageMaker medical image classification pipeline with blue-green endpoint deployment, auto-scaling, and drift detection – zero-downtime model updates with automatic rollback. Provisioned via Terraform + CodePipeline; CloudFront frontend + API Gateway + Lambda inference. Generalized and published by AWS as an official open-source sample (aws-samples, MIT-0).

Amazon Web Services (AWS) Mar 2024 – Jul 2024
Associate Professional Services – DevOps (Internship) Hyderabad, India

- Migrated on-premises application to AWS ECS with blue-green deployment via Terraform, CodePipeline, and CodeDeploy for zero-downtime releases. Established distributed tracing with X-Ray and CloudWatch.

Ikarus-3D May 2023 – Jul 2023
Software Developer Associate (Internship) Mohali, India

- Containerized production application on AWS ECS/ECR with GitHub Actions CI/CD; deployed SonarQube on EC2 + RDS to embed continuous code-quality analysis into the delivery pipeline.

Publications & Talks

- **Published by AWS:** aws-samples/sample-sagemaker-image-classification-mlops (author, MIT-0) – SageMaker MLOps in 86 Terraform files across 4 stacks and 13 modules: clinical quality and Fairlearn fairness gates, hourly PSI drift that starts retraining, canary deploys with auto-rollback. *2026*
- **Published by AWS:** aws-samples/sample-aws-terraform-org-governance (author, MIT-0) – Terraform for AWS Organizations governance on Control Tower: six policy types as 20 policy documents, delegated admin for 15 services. *2026*
- **Talks:** Internal tech talks on Terraform testing and GitOps. Associate Speaker Certification (Renewed 2026); Senior Speaker Certification (2026).
- **Programs:** AWS Professional Services MLOps SME program graduate (2025). AWS technical field community ambassador (AI/ML, DevOps, CloudOps, Agentic AI, Serverless); Anthropic Claude Ambassador (2026). Security content review qualification (2026). AWS Jam content tester, DevOps and GenAI (2026).

Industry Certifications (5)

AWS: Solutions Architect – Associate | Developer – Associate | Machine Learning Engineer – Associate
AWS: AI Practitioner | Cloud Practitioner

Personal projects and opinions are my own and do not represent Amazon or AWS.

Projects

Ledger Sync Jan 2026
React 19, TypeScript, FastAPI, SQLAlchemy, Neon PostgreSQL, Vercel GitHub | Live
◇ Self-hosted personal-finance app: imports .xlsx, .xls and .csv bank statements with SHA-256 transaction IDs so re-imports stay idempotent, then serves 26 analytics pages (Indian fiscal-year tax planning, investments, FIRE, multi-currency) and an AI assistant limited to 15 read-only tool calls. FastAPI on Vercel over Neon PostgreSQL, React on GitHub Pages, with 298 test files across both stacks.

orrery Oct 2026
Agent Skill, Claude Code Plugin, Node.js, ELK, SVG GitHub | Gallery
◇ Agent Skill that turns a YAML spec into animated architecture diagrams with 18,209 official icons (AWS, Azure, Google Cloud, Kubernetes, GitHub, IBM), each cloud pack rebuilt from the vendor's own download and pinned by checksum. ELK auto-layout, a validator whose every finding names the fix, and a review board of 3 reviewer agents; 40 reference diagrams are rendered and checked in CI. Runs in Claude Code, Codex, Cursor, Gemini CLI and Kiro.

Kinfolk Jul 2026
TypeScript, Next.js 16, React 19, Drizzle, Neon Postgres GitHub | Live
◇ Collaborative family graph modelled as a directed acyclic graph, not a tree: each relative keeps their own record and consent links join them, so nothing is overwritten. Live on Vercel + Neon with a production health check; regression tests run on an in-memory PGlite database.

LeetCode Rating Predictor Dec 2023
Python, TensorFlow, NumPy, FastAPI, React, Docker GitHub | Live
◇ Rating model on 15 engineered features. Found and fixed a train/serve skew on participant count that had tripled backtest error (MAE 38.76 to 12.46 on 57 contests). Moved serving to NumPy-only inference pinned by a golden test, cutting the runtime from ~1.8 GB to ~65 MB; a weekly job refreshes the data and re-runs the backtest.

itr-agent Jul 2026
TypeScript, MCP Protocol, Node.js, Zod GitHub
◇ Local-first MCP server for Indian income tax: the LLM interviews and calls 12 tools, while a deterministic engine over a versioned JSON rule pack computes every rupee. 151 tests pin it to published worked examples and the statute, on an Ubuntu and Windows CI matrix.

Other Notable Projects: Claude Cost Optimizer (cache-aware Claude cost model), GitScope (Chrome Web Store extension), Blue-Green AWS Terraform (15 modules, ECS canary), Instagram Likes Leaderboard (browser-console tool), SelfHub (MCP personal-memory server).

Open Source Contributions (11 merged upstream)

- **axios** (109K☆) – AxiosError message now built from AggregateError sub-errors, ending blank dual-stack errors (#11059)
- **Apache Airflow** (47K☆) – template_fields on SalesforceBulkOperator, so its arguments render via Jinja at run time (#63109)
- **lucide** (25K☆) – Fixed a TypeError crash when Vue templates pass the kebab-case icon-node prop (#4512)
- **multer** (12K☆) – Files rejected by fileFilter no longer use up maxCount and fail later valid uploads (#1426)
- **terraform-aws-modules/atlantis** – Fixed custom ECS security group egress rules being silently replaced by allow-all (#446)
- **Also merged:** PrefectHQ/prefect, feast-dev/feast, awslabs/agent-plugins (#132, #212), awslabs/mcp, cloudposse/terraform-aws-tfstate-backend; co-author of the aws-ia/terraform-aws-vpc v4.8.0 fix.

Education

National Institute of Technology Warangal, Telangana Sep 2021 – Jul 2024
Master of Computer Applications (MCA) | NIMCET AIR 208 CGPA: 8.38

Sanskar College of Professional Studies (DAVV University), Indore May 2018 – Aug 2021
Bachelor of Computer Applications (BCA) CGPA: 7.34

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, C++, SQL, HCL, Bash
Systems & Backend: FastAPI, Node.js, Express, gRPC, REST, GraphQL, PostgreSQL, MongoDB, Redis, SQLite, Kafka
ML/AI: PyTorch, TensorFlow, Scikit-learn, SageMaker, Bedrock, Amazon Bedrock AgentCore, MCP (Model Context Protocol), Claude Agent SDK, Feast, LangChain
Cloud & Infra: AWS (ECS, Lambda, SageMaker, API Gateway, CloudFront, IAM, Organizations, Transit Gateway, Network Firewall, AWS Backup, Systems Manager Patch Manager), Terraform, Docker, GitHub Actions, CodePipeline
Frontend: React, Preact, Next.js, Vite, Tailwind, Chrome Extensions (Manifest V3)
CS Fundamentals: Data Structures, Algorithms, System Design, Distributed Systems, OS, DBMS, Computer Networks

Competitive Programming

- **LeetCode Guardian** – 2166 peak rating (top 1.1%), 1300+ problems, 107 contests *Sagargupta1610*
- **Google Kick Start Round E '22** – Rank 1289 *Certificate*
- **GeeksforGeeks** 500+ DSA problems; **HackerRank** 6-Star Problem Solving, 5-Star C++

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